LitWeave: Multimodal Knowledge Content Reconstruction and Presentation for Classic Chinese Literature

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Abstract—Existing digital resources for classic literature are usually stored as independent e-texts, video clips, or encyclopedic materials. The lack of unified organization among texts, videos, characters, locations, and events limits cross-modal narrative understanding and structured browsing. To address this problem, LitWeave, a multimodal knowledge reconstruction and presentation system for classic Chinese literature, is proposed. The system first converts original texts and adapted videos into structured narrative events. Character normalization, location normalization, and unified event representation are then used to construct a multimodal knowledge graph. Based on this graph, a cross-modal event alignment and timeline construction method is designed to map textual scenes and video scenes into a unified event space and generate an ordered narrative timeline. A case study on *Dream of the Red Chamber* shows that LitWeave can organize fragmented text and video resources into structured knowledge involving events, characters, locations, and timelines, providing a feasible approach for multimodal understanding and digital presentation of classic literature.

Keywords—classic Chinese literature; multimodal knowledge reconstruction; knowledge graph; event alignment; timeline construction

I. INTRODUCTION

Classic Chinese literature is an important carrier of traditional Chinese culture. With the development of digitization, related works are now available in various forms, such as electronic texts, online reading platforms, and film or television adaptations. However, existing resources are mostly stored as independent single-modality materials. Texts, images, characters, locations, and plots lack unified organization, which makes it difficult to support structured understanding based on events, character relations, and narrative timelines. Therefore, this paper focuses on multimodal knowledge reconstruction for classic literary resources and aims to organize fragmented textual and video content into browsable, interconnected, and explainable knowledge structures.

Multimodal content understanding provides a technical basis for knowledge unit extraction. CLIP, ALBEF, and BLIP-2 promote the alignment between visual and textual representations, while self-supervised models such as DINO enhance visual feature representation [1~4]. LLaVA-Mini [5] and Qwen2-VL [6] further improve the unified understanding of images, videos, and language through large multimodal models. However, these methods are mainly designed for general visual question answering or content understanding tasks. In long-form literary narratives, they may still suffer from insufficient context modeling, unstable event boundaries, character alias confusion, and biased temporal inference.

Cross-modal retrieval methods can establish similarity associations between texts and video clips. X-CLIP [7] and LoVR [8] improve video-text retrieval and long-video retrieval in multimodal contexts, respectively. However, most retrieval methods remain at the level of embedding-space matching and cannot directly form explainable narrative structures. In contrast, multimodal knowledge graphs provide a unified semantic layer for heterogeneous resources and support the structured organization of entities, events, and relations. Studies such as M2KGRL, text-free multimodal knowledge graph construction, and MMCOMET show that multimodal knowledge graphs can provide a structured foundation for cross-modal knowledge organization and reasoning [9~11]. Therefore, combining large-model-driven multimodal event extraction with knowledge graph construction is a feasible approach for cross-modal knowledge reconstruction of classic literature.

Based on the above analysis, this paper proposes LitWeave, a multimodal knowledge content reconstruction and presentation system for classic Chinese literature. LitWeave takes events as the core unit and organizes texts, videos, characters, locations, and timelines into a unified knowledge graph, supporting event browsing, character trajectory analysis, and location-based narrative presentation.

The main contributions of this paper are as follows:

(1) A unified event representation method is proposed for classic literary texts and film or television content, transforming both modalities into structured narrative events;

(2) A normalization mechanism for characters, locations, and events is constructed to form a multimodal knowledge graph for cross-modal association;

(3) A cross-modal event alignment and timeline construction method is designed, and a LitWeave prototype system is implemented. Its feasibility is verified through a case study on *Dream of the Red Chamber*.

II. METHOD

A. Multimodal Knowledge Content Understanding

To achieve unified knowledge reconstruction of classic literary texts and film or television content, the original inputs of the two modalities must first be transformed into a consistent narrative event representation. Let the input sequence be $x^{(m)}$, where $m \in \{\text{video},\text{text}\}$. The task of multimodal knowledge content understanding can be formulated as:

$$F:x^{(m)} \rightarrow E^{(m)}=\{e_1,e_2,...,e_n\}. \quad (1)$$

Here, e_i denotes a narrative event unit rather than a mechanical segmentation result based on shots or paragraphs.

On the text side, chapter-level texts are used as the basic input units. When the input is too long, a heuristic segmentation strategy based on length balance and boundary quality is adopted to obtain more natural text segments. On the video side, coarse-grained segmentation is first conducted, and event extraction is then performed within each segment. Under a unified structured prompt constraint, the model simultaneously performs event segmentation and event attribute extraction, and outputs a unified JSON format containing:

$$e_i=\left(a_i,c_i,l_i,t_i^h,t_i^e,r_i,s_i\right), \quad (2)$$

These fields correspond to the action, characters, location, coarse-grained temporal clue, explicit time expression, event relation, and event summary, respectively. For text events, the field source_text is further retained as textual evidence. The results show that this method can steadily transform text and video into a unified event representation, providing reliable input for subsequent knowledge graph construction.

B. Multimodal Knowledge Graph Construction

After event extraction, text and video are transformed into structured events with homogeneous fields. However, these results cannot be directly written into a unified knowledge graph. The reason is that the same character may appear in the form of a full name, abbreviation, alias, or relational address, while the same location may be expressed as a standard place name, a local subspace, or a generalized scene label. Direct graph construction would lead to fragmented entity nodes and inconsistent event organization. Therefore, in the knowledge graph construction stage, this study performs character normalization, location normalization, and unified event representation in sequence, so that multimodal events can be organized into the same knowledge space.

For character normalization, an external character knowledge base is constructed to reduce the fragmentation of character nodes, and standardized character normalization is performed. Let M_c denote the original character mention set, and let C denote the manually curated standard character set. The character normalization task is defined as:

$$f_c:M_c \rightarrow C \cup \{\text{generic},\text{visual_noise},\text{unknown}\}. \quad (3)$$

Specifically, character mentions are first classified into four types: canonical, alias, generic, and visual_noise. Then, only alias-type mentions are submitted to a closed-set LLM mapping process, where each mention is mapped to an existing standard character name or marked as unknown. The final character knowledge base is represented as:

$$K=\{(c_i,A_i)\}. \quad (4)$$

Here, c_i denotes a standard character name, and A_i denotes its alias set. This knowledge base provides a unified entity layer for subsequent multimodal knowledge graph construction.

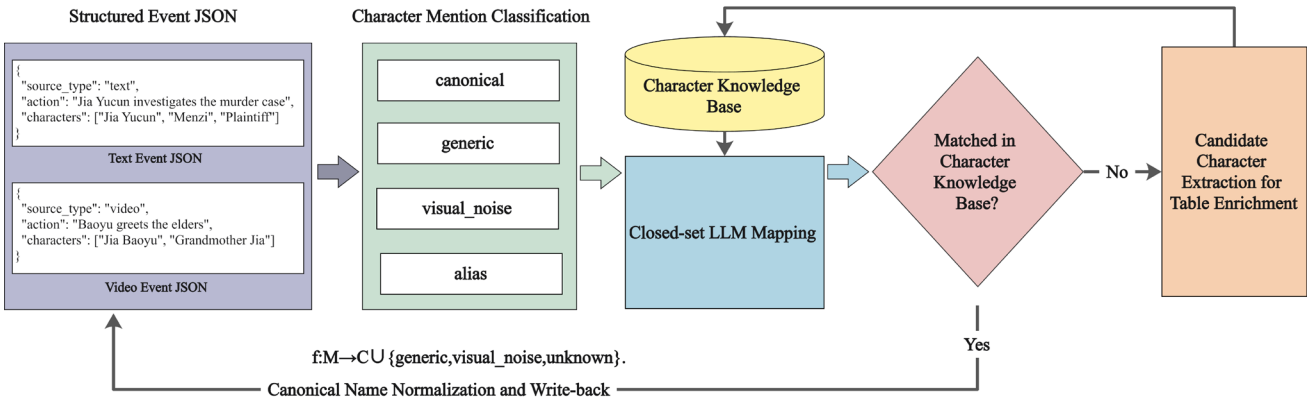


Fig. 1. Character Normalization Framework Based on an External Knowledge Base

For location normalization, a closed-set mapping strategy similar to character normalization is adopted. Let M_l denote the original location mention set, and let L denote the standard location set. The location normalization task is defined as:

$$f_l: M_l \rightarrow L \cup \{\text{generic}, \text{unknown}\}. \quad (5)$$

Location mentions are classified into five types: canonical, generic, descriptive, subspace, and alias. Only descriptive, subspace, and alias mentions are submitted to a closed-set LLM mapping process, where they are mapped to standard locations or marked as unknown. This strategy merges fine-grained spatial expressions into stable location nodes and avoids generating incorrect locations outside the closed set.

For unified event representation, unified event objects are further constructed after character and location normalization. Let $E^{(\text{text})}$ and $E^{(\text{video})}$ denote the text event set and video event set, respectively. The unified event set is defined as:

$$E = \text{Norm}\left(E^{(\text{text})} \cup E^{(\text{video})}\right). \quad (6)$$

Each unified event retains core fields such as action, normalized character set, normalized location, temporal clue, event relation, and event summary. A unique event identifier and an event comparison text are also generated. In this way, text events and video events are organized into the same index space, providing standard input for subsequent cross-modal event alignment and timeline construction.

C. Association Algorithm Based on the Multimodal Knowledge Graph

After constructing the multimodal knowledge graph, the system further organizes unified events, characters, locations, and temporal information into a browsable narrative structure. Since characters and locations are attached to event nodes, this study takes events as the core unit of cross-modal association and focuses on two tasks. The first is to establish event alignment between textual scenes and video scenes. The second is to determine the position of aligned events on the overall narrative timeline. Based on the unified event graph, character trajectories and location timelines can be further derived.

For cross-modal event alignment, this study addresses the differences between classic literary texts and film or television narratives in event granularity, expression form, and scene segmentation. Directly matching atomic events may cause mismatches. Therefore, this study adopts a hierarchical alignment strategy consisting of atomic events, scene events, canonical events, and cross-modal links. First, adjacent atomic events are aggregated into semantically more stable scene events. Then, a canonical event knowledge base is introduced as a semantic mediator to map textual scenes and video scenes into a unified event space.

Let s denote a scene event and k denote a canonical event. The matching score between a scene event and the knowledge base is defined as:

$$\begin{aligned}\text{Score}(s,k) = {} & \alpha S_{\text{name}}(s,k) + \beta S_{\text{detail}}(s,k) \\ & + \gamma J_{\text{char}}(s,k) + \delta J_{\text{loc}}(s,k) + \text{Bonus}(k)\end{aligned} \quad (7)$$

where S_{name} and S_{detail} denote title-level and detail-level semantic similarity, respectively. J_{char} and J_{loc} denote the Jaccard similarity of character sets and location sets, respectively. $\text{Bonus}(k)$ denotes the bias term for cross-modal anchor events. The canonical event with the highest score above the threshold is selected as the mapping result of the scene event.

In the cross-modal linking stage, the system computes link scores only for textual scenes s_t and video scenes s_v that share the same canonical event. The link score is defined as:

$$\begin{aligned}\text{LinkScore}(s_t, s_v) = {} & 0.45 S_{\text{textKB}} + 0.35 S_{\text{videoKB}} \\ & + 0.12 J_{\text{char}}(s_t, s_v) + 0.08 J_{\text{loc}}(s_t, s_v).\end{aligned} \quad (8)$$

where S_{textKB} and $S_{videoKB}$ denote the matching scores of the textual scene and the video scene to the canonical event, respectively. When the link score exceeds the threshold, a text-video cross-modal event link is generated. For boundary cases, an LLM-based closed-set review is introduced to improve the stability of complex scene discrimination.

For event timeline construction, cross-modal event alignment establishes the correspondence between text and video, but the position of each event in the overall narrative still needs to be determined. Since classic literary narratives often contain flashbacks, interludes, and implicit temporal clues, this study introduces an external timeline reference database to locate aligned events. Let e denote a cross-modal event and t denote a timeline node. The timeline grounding score is defined as:

$$\begin{aligned}Score_{timeline}(e,t) = {} & \alpha S_{name}(e,t) \\ & + \beta S_{summary}(e,t) + \gamma B_{kb}(e,t),\end{aligned} \quad (9)$$

where S_{name} denotes the similarity between the event name and the timeline event name, $S_{summary}$ denotes the semantic similarity between the fused event summary and the timeline entry, and B_{kb} denotes the coverage bonus of the canonical event name. Candidate timeline nodes are obtained according to this score. Then, LLM-based closed-set judgment and necessary manual correction are used to determine the final temporal position of the event.

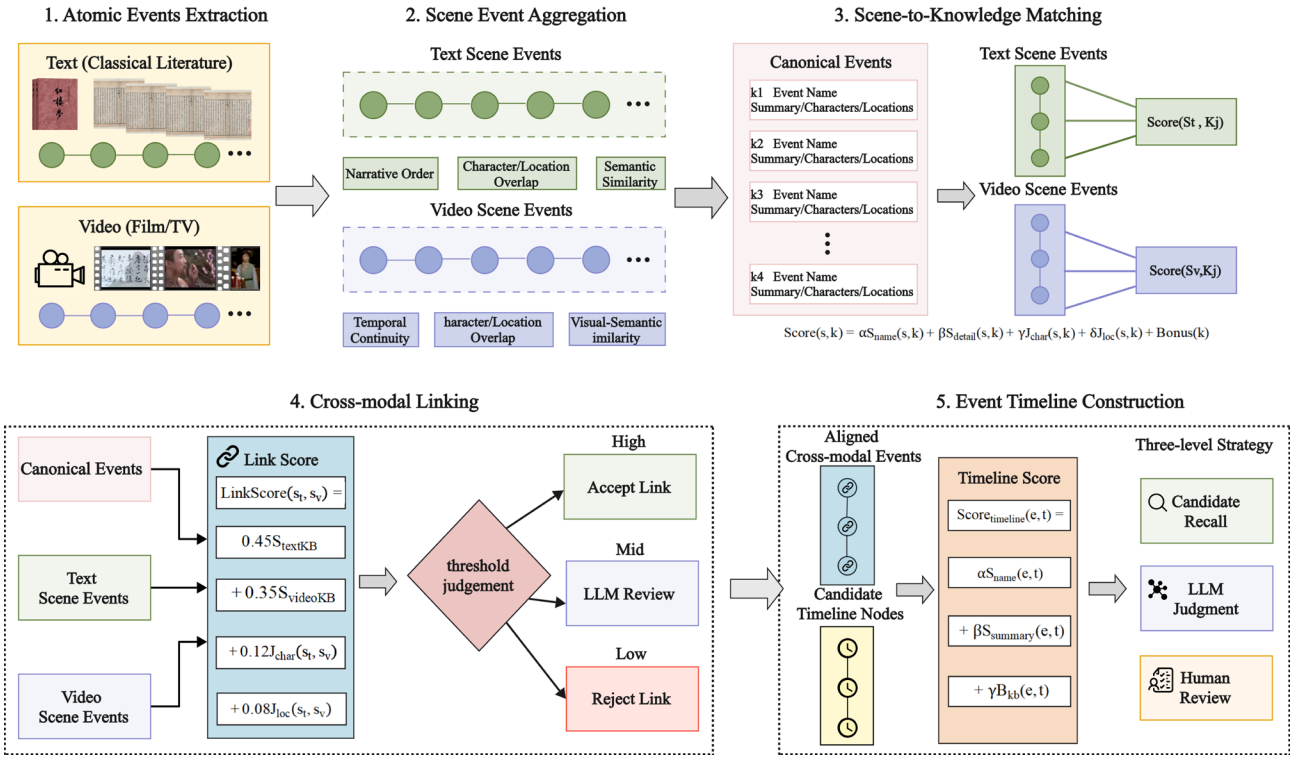


Fig. 2. Canonical-event-based Cross-modal Event Alignment and Timeline Construction Workflow

Through the above process, the system obtains a structured narrative graph with canonical event labels, text-video corresponding segments, and temporal positions, as shown in Fig. 2. Characters and locations are no longer matched independently across modalities. Instead, they are derived from the unified event graph. The system aggregates related events according to the character and location information in each event and arranges them in temporal order, thereby forming character timelines and location timelines. This design ensures the consistency of events, characters, and locations within the same knowledge space.

III. System Design

A. System Architecture

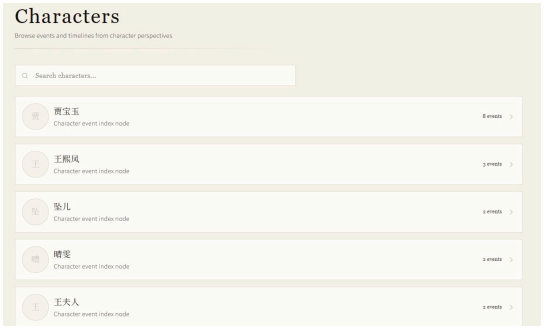
LitWeave adopts a B/S architecture with front-end/back-end separation. The overall system consists of a data layer, a knowledge processing layer, a service layer, and a presentation layer. The data layer stores original texts, film or television clips, structured events, the character knowledge base, the location knowledge base, and the timeline reference database. The knowledge processing layer is responsible for event extraction, entity normalization, event alignment, and timeline construction. The service layer provides RESTful APIs for the front end to query events, characters, locations, and multimodal content. The presentation layer organizes knowledge browsing around three entry points: events, characters, and locations. This architecture enables text and video resources to be associated and presented within a unified event graph.

B. Functional Module Design

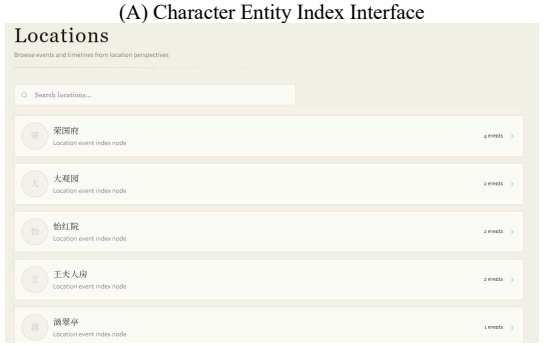
The front end of the system mainly includes the event module, character module, and location module. The event module presents a unified event timeline sorted by time, where users can view event names, time, characters, locations, and summaries, and enter the detail page to view corresponding text and video segments. The character module derives character indexes from the event graph and displays the event timeline in which each character participates. The location module derives location-based indexes and presents event sequences occurring at the same location. All three modules point to the unified event detail page, where textual segments, video segments, and structured knowledge are presented together.

C. Functional Demonstration

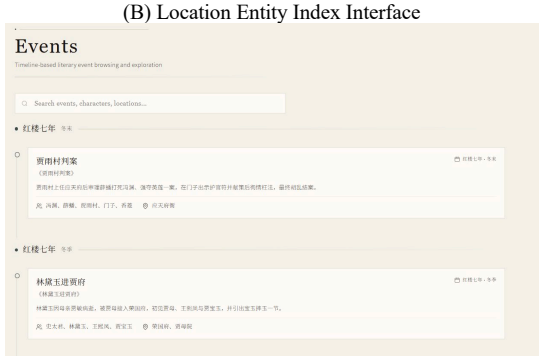
A prototype system is constructed using *Dream of the Red Chamber* as a case study. The original text and the adapted television series are processed through event extraction, character and location normalization, cross-modal alignment, and timeline grounding, and are then organized into unified event master data. The system interface provides a global event timeline, character event timelines, location event timelines, and event detail pages. Users can enter an event node from any entrance and view its corresponding content in both text and video. Fig. 3 shows the prototype interface of LitWeave, including the event, character, location, and event-detail modules, thereby verifying the feasibility of structured reconstruction and interactive presentation of multimodal content in classic literature.



(A) Character Entity Index Interface



(B) Location Entity Index Interface



(C) Chronological Event Timeline Interface



(D) Multimodal Event Presentation Interface

Fig. 3. Interface Demonstration of the LitWeave Prototype System

IV. CONCLUSION

This paper proposes LitWeave, a multimodal knowledge content reconstruction and presentation system for classic Chinese literature. The system takes events as the core units, transforms original texts and film or television content into structured events, and constructs a browsable multimodal narrative knowledge graph through character normalization, location normalization, canonical event mapping, and timeline construction. The system implementation based on *Dream of the Red Chamber* shows that LitWeave can organize fragmented text and video resources into a knowledge structure involving events, characters, locations, and timelines. It provides a feasible approach for digital understanding and presentation of classic literature. In future work, we will expand the scale of text and video data, improve the quality of cross-modal alignment, and introduce richer interactive graph visualization functions.

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